ELEVATE LABS CYBER SECURITY INTERNSHIP

Task 1
Local Network Port Scanning

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Objective

The objective of this task was to perform basic network reconnaissance to discover active hosts and open ports within a local authorized network. This helps in understanding network exposure and identifying potential security risks associated with exposed services.

Methodology & Tools Used

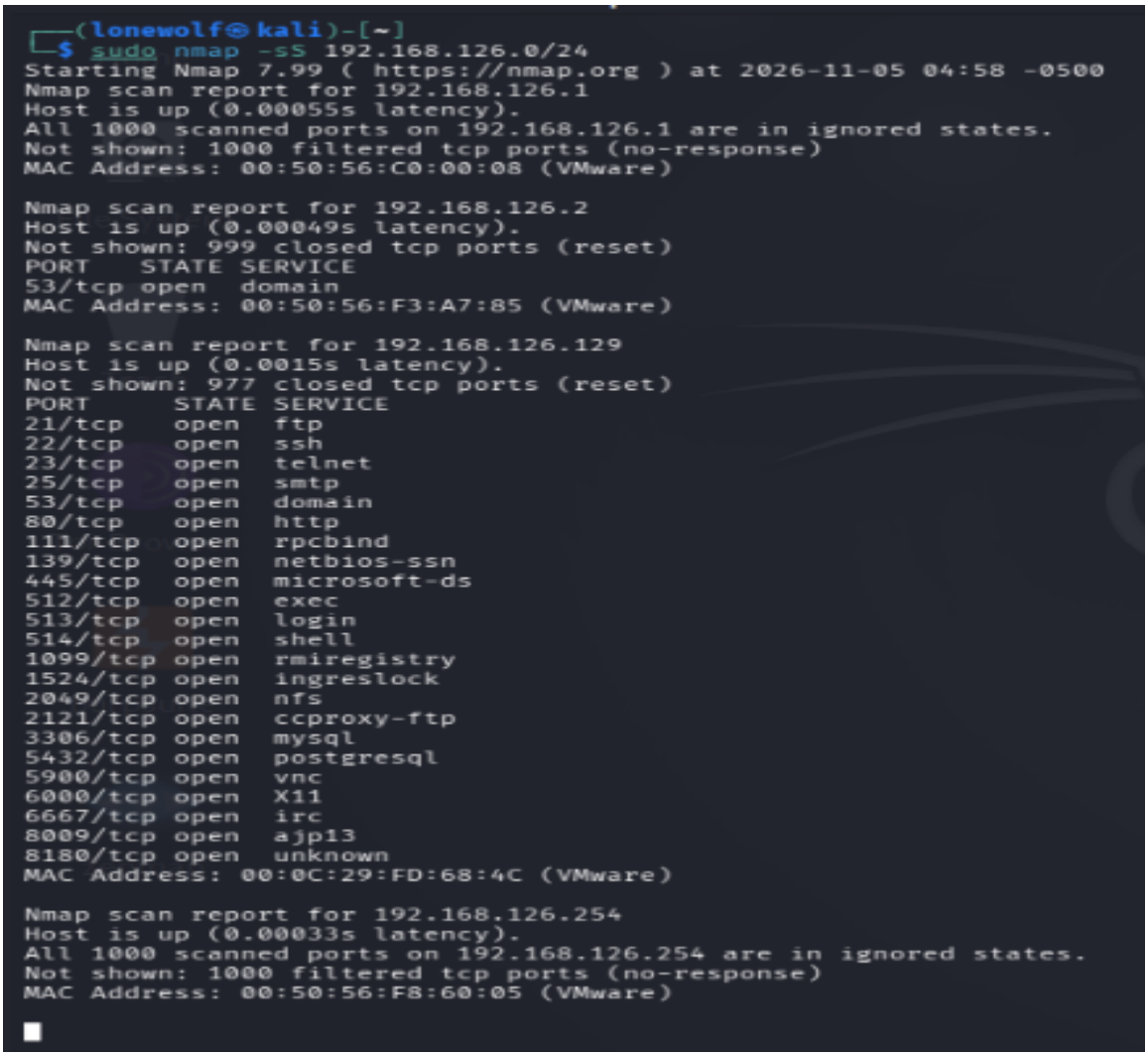
- **Tool:** Nmap (Network Mapper) v7.99
- **Scan Type:** TCP SYN Scan (-sS). This was chosen because it is fast, stealthy, and does not complete the full TCP handshake, making it less likely to be logged by the target's applications.
- **Command Executed:** `sudo nmap -sS 192.168.126.0/24 -oN network_scan.txt`

Scan Results & Findings

The scan discovered multiple live hosts on the network, including the VMware NAT gateway and the intended target machine (Metasploitable 2).

Key Findings on Target (192.168.126.129):
The target machine (Metasploitable 2) had numerous exposed ports. The most critical ones include:

- **Port 21 (FTP):** File Transfer Protocol.
- **Port 22 (SSH):** Secure Shell.
- **Port 23 (Telnet):** Remote login (Unencrypted).
- **Port 80 (HTTP):** Web server.
- **Port 445 (SMB):** Server Message Block (File sharing).



Service Analysis & Security Risks

Leaving ports open exposes the services running behind them to the network. Here is an analysis of the risks:

- **Port 21 (FTP):** FTP transmits data, including usernames and passwords, in clear text. An attacker on the same network can use Wireshark to sniff these credentials. **Risk: High.**
- **Port 23 (Telnet):** Similar to FTP, Telnet is completely unencrypted. It should be replaced with SSH (Port 22). **Risk: High.**
- **Port 80 (HTTP):** Unencrypted web traffic. If users log into this web server, their session cookies and passwords can be intercepted. **Risk: Medium.**

- **Port 445 (SMB):** SMB file sharing exposed to the network is highly dangerous. It is the primary attack vector for devastating ransomware like WannaCry and NotPetya. **Risk: Critical.**

Conclusion & Key Learnings

This task demonstrated the importance of network reconnaissance. By using Nmap, I was able to quickly map the network and identify high-risk services running on the target machine.

Key Learning: An open port is not inherently a vulnerability, but an unnecessary open port significantly increases the attack surface. Best practices dictate that administrators should close any port that is not strictly required for business operations (Principle of Least Privilege) and replace insecure protocols (like FTP and Telnet) with encrypted alternatives (like SFTP and SSH).